MA1522 Linear Algebra for Computing

Lesson 17: Diagonalization and Orthogonal Diagonalization

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson17` (e.g. `A1234567Z_lesson17.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 17**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

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Example 1

True or false? Let A be a square matrix and $\lambda_1 \neq \lambda_2$ be two distinct eigenvalues of A. Let S_1 and S_2 be bases of the eigenspaces E_{λ_1} and E_{λ_2} respectively. Then

1. $S_1 \cup S_2$ is linearly independent.
2. S_1 is orthogonal to S_2.

1. **True**: eigenspaces associated to distinct eigenvalues are linearly independent (Section 6.2).

2. False. Let $M = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$, with eigenvalues 0 and 1. Then

$$E_0 = \text{span}\left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right\}, \qquad E_1 = \text{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}, \qquad \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \neq 0.$$

Section 6.2 Diagonalization

Let A be a square matrix of order n.

Definition

A is **diagonalizable** if there exists an invertible matrix P such that $P^{-1}AP = D$ is a diagonal matrix, or equivalently, $A = PDP^{-1}$.

Theorem

A is diagonalizable if and only if A has n linearly independent eigenvectors. In that case

$$P = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \end{pmatrix}, \qquad D = \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n),$$

where $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is a basis for $\mathbb{R}^n$ of eigenvectors of A and $A\mathbf{u}_i = \mu_i \mathbf{u}_i$.

Section 6.2 Equivalent Statements for Diagonalizability

Theorem

For every eigenvalue λ of A, $\quad 1 \leq \dim(E_\lambda) \leq r_\lambda$.

Theorem

A is diagonalizable if and only if

1. the characteristic polynomial splits into linear factors, $\det(xI - A) = (x - \lambda_1)^{r_{\lambda_1}} \cdots (x - \lambda_k)^{r_{\lambda_k}}$, and
2. $\dim(E_{\lambda_i}) = r_{\lambda_i}$ for every eigenvalue λ_i.

Section 6.2 Algorithm for Diagonalization

Let A be a square matrix of order n.

1. Compute and factorise $\det(xI - A) = (x - \lambda_1)^{r_{\lambda_1}} \cdots (x - \lambda_k)^{r_{\lambda_k}}$. If it does not split into linear factors, stop.

2. For each eigenvalue λ_i, find a basis S_{λ_i} for E_{λ_i}, the solution space of $(\lambda_i I - A)\mathbf{x} = \mathbf{0}$. If $\dim(E_{\lambda_i}) < r_{\lambda_i}$, stop.

3. $S = S_{\lambda_1} \cup \cdots \cup S_{\lambda_k} = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is a basis for $\mathbb{R}^n$ of eigenvectors of A.

4. Let $P = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{pmatrix}$ and $D = \operatorname{diag}(\mu_1, \ldots, \mu_n)$, where $A\mathbf{u}_i = \mu_i \mathbf{u}_i$. Then $A = PDP^{-1}$.

Question 1

Let

$$S = \begin{pmatrix} -4 & 1 & -2 \\ 1 & -4 & -2 \\ -2 & -2 & -1 \end{pmatrix}, \qquad \det(xI - S) = (x+5)^2(x-1).$$

1. For each eigenvalue λ of S, find a basis for E_λ.

2. Is S diagonalizable? If so, find an invertible matrix P and a diagonal matrix D such that $S = PDP^{-1}$. Check your answer.

Question 1 — solution: part 1

1. $E_{-5} = \text{Null}(-5I - S)$:

$$-5I - S = \begin{pmatrix} -1 & -1 & 2 \\ -1 & -1 & 2 \\ 2 & 2 & -4 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad E_{-5} = \text{span}\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

$E_1 = \text{Null}(I - S)$:

$$I - S = \begin{pmatrix} 5 & -1 & 2 \\ -1 & 5 & 2 \\ 2 & 2 & 2 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}.$$

With $x_3 = s$: $x_1 = x_2 = -\frac{1}{2}s$. Taking $s = -2$, $E_1 = \text{span}\{(1, 1, -2)^T\}$.

Question 1 — solution: part 2

2. Is S diagonalizable? If so, find P invertible and D diagonal with $S = PDP^{-1}$. Check your answer.

$\det(xI - S)$ splits, $\dim(E_{-5}) = 2 = r_{-5}$ and $\dim(E_1) = 1 = r_1$, so S is diagonalizable. Put the eigenvectors in P, and their eigenvalues in D in the same order:

$$P = \begin{pmatrix} -1 & 2 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix}, \qquad D = \begin{pmatrix} -5 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Check $SP = PD$ instead of computing P^{-1}:

$$SP = \begin{pmatrix} 5 & -10 & 1 \\ -5 & 0 & 1 \\ 0 & -5 & -2 \end{pmatrix} = PD. \checkmark$$

□

Question 2

Let a be a real number, and

$$B_a = \begin{pmatrix} 2 & 1 & a \\ 0 & -2 & 4 \\ 0 & 0 & 2 \end{pmatrix}.$$

1. Find all values of a for which B_a is diagonalizable.

2. For each such a, find an invertible matrix P and a diagonal matrix D such that $B_a = PDP^{-1}$.

Question 2 — solution: part 1

1. B_a is upper triangular, so $\det(xI - B_a) = (x-2)^2(x+2)$: eigenvalues 2, with $r_2 = 2$, and -2, with $r_{-2} = 1$, for every a.

Since $1 \leq \dim(E_{-2}) \leq r_{-2} = 1$, $\dim(E_{-2}) = 1$ for every a. So B_a is diagonalizable if and only if $\dim(E_2) = 2$, i.e. $\operatorname{rank}(2I - B_a) = 1$:

$$2I - B_a = \begin{pmatrix} 0 & -1 & -a \\ 0 & 4 & -4 \\ 0 & 0 & 0 \end{pmatrix}.$$

The first two rows are linearly dependent exactly when $(-1, -a) = -\frac{1}{4}(4, -4) = (-1, 1)$, i.e. $a = -1$. For $a \neq -1$ the rank is 2 and $\dim(E_2) = 1 < 2$.

So B_a is diagonalizable if and only if $a = -1$.

Question 2 — solution: part 2

2. Take $a = -1$. For E_2, the rows of $2I - B_{-1}$ are multiples of $(0, 1, -1)$, so $x_2 = x_3$ and x_1 is free:

$$E_2 = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

For E_{-2}, read off the rows of $-2I - B_{-1}$ without reducing to RREF:

$$-2I - B_{-1} = \begin{pmatrix} -4 & -1 & 1 \\ 0 & 0 & -4 \\ 0 & 0 & -4 \end{pmatrix} : \quad x_3 = 0,\ x_2 = -4x_1, \qquad E_{-2} = \text{span}\left\{ \begin{pmatrix} 1 \\ -4 \\ 0 \end{pmatrix} \right\}.$$

$$P = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -4 \\ 0 & 1 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \qquad B_{-1}P = \begin{pmatrix} 2 & 0 & -2 \\ 0 & 2 & 8 \\ 0 & 2 & 0 \end{pmatrix} = PD. \checkmark$$

□

Homework

For each matrix below, determine whether it is diagonalizable. If it is, find an invertible matrix P and a diagonal matrix D such that it equals PDP^{-1}.

$$H_1 = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix}, \qquad H_2 = \begin{pmatrix} 3 & 0 & 0 \\ 1 & -2 & -3 \\ 4 & 0 & 1 \end{pmatrix}, \qquad H_3 = \begin{pmatrix} -1 & -4 & 1 \\ 0 & 3 & -1 \\ 0 & 1 & 1 \end{pmatrix},$$

$$H_4 = \begin{pmatrix} -1 & 3 & -3 & -3 \\ 0 & -1 & -3 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & -3 & -3 & 2 \end{pmatrix}, \qquad H_5 = \begin{pmatrix} 2 & -1 & -3 & 2 \\ 0 & 3 & 0 & 0 \\ 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 \end{pmatrix}.$$

The eigenvalues of $H_2, \ldots, H_5$ are integers.

Section 6.3 Orthogonally Diagonalizable

Definition

A square matrix A of order n is **orthogonally diagonalizable** if $A = PDP^T$ for some orthogonal matrix P (that is, $P^T = P^{-1}$) and diagonal matrix D.

Theorem (Spectral theorem)

A is orthogonally diagonalizable if and only if A is symmetric. Equivalently, there is an orthonormal basis of $\mathbb{R}^n$ of eigenvectors of A.

Theorem

If A is symmetric, $\lambda_1 \neq \lambda_2$ are eigenvalues of A and $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$, then $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$: the eigenspaces of a symmetric matrix are orthogonal to each other.

Section 6.3 Algorithm for Orthogonal Diagonalization

Let A be a symmetric matrix of order n.

1. Compute $\det(xI - A) = (x - \lambda_1)^{r_{\lambda_1}} \cdots (x - \lambda_k)^{r_{\lambda_k}}$.

2. For each eigenvalue λ_i, find a basis S_{λ_i} for E_{λ_i}.

3. Apply the Gram–Schmidt process to each S_{λ_i} to get an orthonormal basis T_{λ_i} of E_{λ_i}. Then $T = T_{\lambda_1} \cup \cdots \cup T_{\lambda_k} = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is an orthonormal basis for $\mathbb{R}^n$.

4. Let $P = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{pmatrix}$ and $D = \operatorname{diag}(\mu_1, \ldots, \mu_n)$, where $A\mathbf{u}_i = \mu_i \mathbf{u}_i$. Then P is orthogonal and $A = PDP^T$.

Question 3

Let S be the matrix in Question 1,

$$S = \begin{pmatrix} -4 & 1 & -2 \\ 1 & -4 & -2 \\ -2 & -2 & -1 \end{pmatrix}.$$

Orthogonally diagonalize S: find an orthogonal matrix P and a diagonal matrix D such that $S = PDP^T$.

Question 3 — solution

From Question 1, $E_{-5} = \operatorname{span}\{(-1,1,0)^T, (2,0,1)^T\}$ and $E_1 = \operatorname{span}\{(1,1,-2)^T\}$; E_1 is orthogonal to E_{-5}. Apply Gram–Schmidt inside E_{-5}, with $\mathbf{w}_1 = (-1,1,0)^T$, $\mathbf{w}_2 = (2,0,1)^T$:

$$\mathbf{v}_1 = \mathbf{w}_1, \qquad \mathbf{v}_2 = \mathbf{w}_2 - \frac{\mathbf{w}_2 \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} - \frac{-2}{2}\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Normalise $\mathbf{v}_1$, $\mathbf{v}_2$ and $(1,1,-2)^T$, and put them in P in the order of D:

$$P = \begin{pmatrix} -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & -2/\sqrt{6} \end{pmatrix}, \qquad D = \begin{pmatrix} -5 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

□

Next lesson

Next lesson (**Tue 27 Oct**, Sections 6.4 & 6.5), I will focus on

1. powers of diagonalizable matrices, and Markov chains;

2. the **singular value decomposition**.

Please watch the video clips beforehand.

Homework — answers

H_1: $\det(xI - H_1) = x^2 - 2x + 5 = (x-1)^2 + 4$ has no real root, so it does not split: not diagonalizable.

H_2: $(x-3)(x-1)(x+2)$, three distinct eigenvalues: diagonalizable. $E_3 = \operatorname{span}\{(1,-1,2)^T\}$, $E_1 = \operatorname{span}\{(0,-1,1)^T\}$, $E_{-2} = \operatorname{span}\{(0,1,0)^T\}$; $P = \begin{pmatrix} 1 & 0 & 0 \\ -1 & -1 & 1 \\ 2 & 1 & 0 \end{pmatrix}$, $D = \operatorname{diag}(3,1,-2)$.

H_3: $(x-2)^2(x+1)$. $E_2 = \operatorname{span}\{(-1,1,1)^T\}$, so $\dim(E_2) = 1 < 2$: not diagonalizable.
($E_{-1} = \operatorname{span}\{(1,0,0)^T\}$.)

H_4: $(x-2)^2(x+1)^2$. $E_2 = \operatorname{span}\{(-2,-1,1,0)^T, (-1,0,0,1)^T\}$, $E_{-1} = \operatorname{span}\{(1,0,0,0)^T, (0,1,0,1)^T\}$: diagonalizable,

$$P = \begin{pmatrix} -2 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad D = \operatorname{diag}(2,2,-1,-1).$$

H_5: $(x-3)^3(x-1)$. $E_3 = \operatorname{span}\{(-1,1,0,0)^T, (-1,0,1,1)^T\}$, so $\dim(E_3) = 2 < 3$: not diagonalizable.
($E_1 = \operatorname{span}\{(-2,0,0,1)^T\}$.)